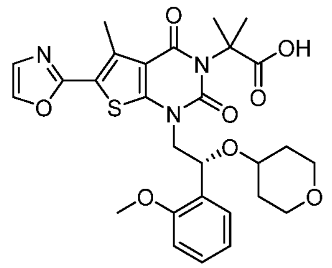
What is claimed is:

1. A choline salt or co-crystal of Compound I:



I

having a crystalline form characterized by an X-ray powder diffractogram comprising peaks at 5.0, 7.8, and 9.4 °2θ ± 0.2 °2θ, as determined on a diffractometer using Cu-Kα radiation at a wavelength of 1.5406 Å.

2. The choline salt or co-crystal of Compound I of claim 1, wherein the diffractogram further comprises peaks at 17.6, 21.3, and 23.9 ° 2θ ± 0.2 °2θ.

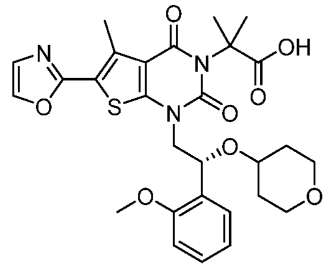
3. The choline salt or co-crystal of Compound I of claim 1 or claim 2, wherein the diffractogram further comprises peaks at 11.0, 16.4, and 20.5 ° 2θ ± 0.2 °2θ.

4. The choline salt or co-crystal of Compound I of any one of claims 1-3, wherein the diffractogram is substantially as shown in Figure 1.

5. The choline salt or co-crystal of Compound I of any one of claims 1-4, characterized by a differential scanning calorimetry (DSC) curve that comprises an endotherm at about 73 °C and an endotherm at about 195 °C.

6. The choline salt or co-crystal of Compound I of any one of claims 1-5, wherein the DSC curve is substantially as shown in Figure 2.

7. A diethylamine salt or co-crystal of Compound I:



I

having a crystalline form characterized by an X-ray powder diffractogram comprising peaks at 6.5, 8.5, and 21.6 ° 2θ ± 0.2 °2θ, as determined on a diffractometer using Cu-Kα radiation at a wavelength of 1.5406 Å.

8. The diethylamine salt or co-crystal of Compound I of claim 7, wherein the diffractogram further comprises peaks at 9.7, 11.5, and 12.0 ° 2θ ± 0.2 °2θ.

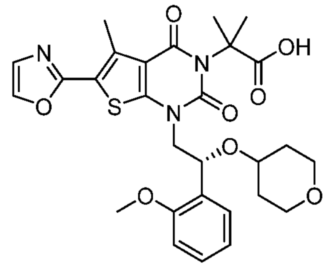
9. The diethylamine salt or co-crystal of Compound I of claim 7 or claim 8, wherein the diffractogram further comprises peaks at 21.1, and 22.8, 27.7 ° 2θ ± 0.2 °2θ.

10. The diethylamine salt or co-crystal of Compound I of any one of claims 7-9, wherein the diffractogram is substantially as shown in Figure 4.

11. The diethylamine salt or co-crystal of Compound I of any one of claims 7-10, characterized by a differential scanning calorimetry (DSC) curve that comprises an endotherm at about 135 °C and an endotherm at about 171 °C.

12. The diethylamine salt or co-crystal of Compound I of any one of claims 7-11, wherein the DSC curve is substantially as shown in Figure 6.

13. A *N,N*-dibenzylethylenediamine salt or co-crystal of Compound I:



I

having a crystalline form characterized by an X-ray powder diffractogram comprising peaks at 4.7, 5.6, and 14.0 ° 2θ ± 0.2 °2θ, as determined on a diffractometer using Cu-Kα radiation at a wavelength of 1.5406 Å.

14. The *N,N*-dibenzylethylenediamine salt or co-crystal of Compound I of claim 13, wherein the diffractogram further comprises peaks at 7.0, 16.9, and 19.6 ° 2θ ± 0.2 °2θ.

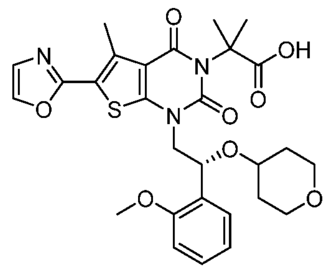
15. The *N,N*-dibenzylethylenediamine salt or co-crystal of Compound I of claim 13 or claim 14, wherein the diffractogram further comprises peaks at 8.7, 10.7, and 17.8 ° 2θ ± 0.2 °2θ.

16. The *N,N*-dibenzylethylenediamine salt or co-crystal of Compound I of any one of claims 13-15, wherein the diffractogram is substantially as shown in Figure 8.

17. The *N,N*-dibenzylethylenediamine salt or co-crystal of Compound I of any one of claims 13-16, characterized by a differential scanning calorimetry (DSC) curve that comprises an endotherm at about 81 °C.

18. The *N,N*-dibenzylethylenediamine salt or co-crystal of Compound I of any one of claims 13-17, wherein the DSC curve is substantially as shown in Figure 9.

19. An ethanolamine salt or co-crystal of Compound I:



I

having a crystalline form characterized by an X-ray powder diffractogram comprising peaks at 5.4, 7.2, and 10.0 ° 2θ ± 0.2 °2θ, as determined on a diffractometer using Cu-Kα radiation at a wavelength of 1.5406 Å.

20. The ethanolamine salt or co-crystal of Compound I of claim 19 wherein the diffractogram further comprises peaks at 15.4, 19.1, and 20.7 ° 2θ ± 0.2 °2θ.

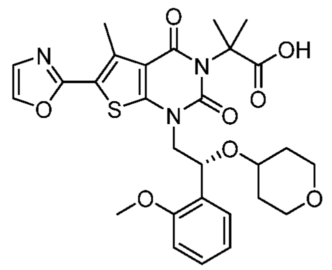
21. The ethanolamine salt or co-crystal of Compound I of claim 19 or claim 20, wherein the diffractogram further comprises peaks at 21.6, 23.4, and 28.3 ° 2θ ± 0.2 °2θ.

22. The ethanolamine salt or co-crystal of Compound I of any one of claims 19-21, wherein the diffractogram is substantially as shown in Figure 11.

23. The ethanolamine salt or co-crystal of Compound I of any one of claims 19-22, characterized by a differential scanning calorimetry (DSC) curve that comprises an endotherm at about 22 °C and an endotherm at about 133 °C.

24. The ethanolamine salt or co-crystal of Compound I of any one of claims 19-23, wherein the DSC curve is substantially as shown in Figure 12.

25. A crystalline form of Compound I:



I

(Compound I Form IX) characterized by an X-ray powder diffractogram comprising peaks at 7.2, 7.8, and 14.8 ° 2θ ± 0.2 °2θ, as determined on a diffractometer using Cu-Kα radiation at a wavelength of 1.5406 Å.

26. The crystalline form of claim 25, wherein the diffractogram further comprises peaks at 19.8, 23.1, and 25.5 ° 2θ ± 0.2 °2θ.

27. The crystalline form of claim 25 or claim 26, wherein the diffractogram further comprises peaks at 16.8, 20.8, and 22.6 ° 2θ ± 0.2 °2θ.

28. The crystalline form of any one of claims 25-27, wherein the diffractogram is substantially as shown in Figure 14.

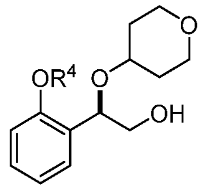
29. The crystalline form of any one of claims 25-28, characterized by a differential scanning calorimetry (DSC) curve that comprises an endotherm at about 85 °C.

30. The crystalline form of any one of claims 25-29, wherein the DSC curve is substantially as shown in Figure 15.

31. A pharmaceutical composition comprising the salt or co-crystal or crytalline form of Compound I of any one of claims 1-30 and a pharmaceutically acceptable carrier, adjuvant or diluent.

32. A method of treating an ACC-mediated disorder comprising administering to a patient in need thereof a therapeutically effective amount of the salt or co-crystal or crystalline form of Compound I of any one of claims 1-30 or a pharmaceutical composition of claim 31.

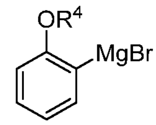
33. A method for preparing compound of formula (J):



(J),

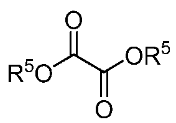
comprising the steps of:

(a) contacting a compound of formula (R):



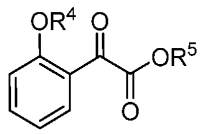
(R)

with a compound of formula (S):



(S)

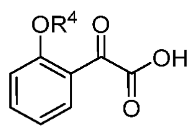
under conditions sufficient to form a compound of formula (P):



(P)

or a solvate or a hydrate thereof,

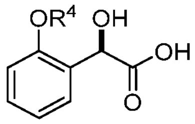
(b) contacting a compound of formula (P), or a solvate or a hydrate thereof, with a base under conditions sufficient to form a compound of formula (O):



(O)

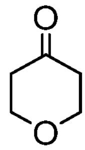
or a salt, a solvate, or a hydrate thereof,

(c) contacting a compound of formula (O), or a salt, a solvate, or a hydrate thereof, with a reductant and a catalyst under conditions sufficient to form a compound of formula (N):



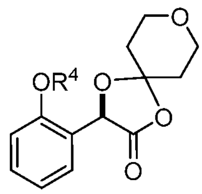
(N);

(d) contacting a compound of formula (N) with a compound of formula (M):



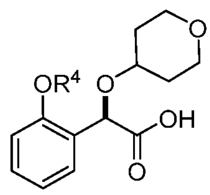
(M)

under conditions sufficient to form a compound of formula (L):



(L);

(e) contacting a compound of formula (L) with a reductant under conditions sufficient to form a compound of formula (K):

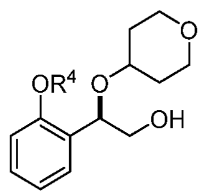


(K);

and (f) contacting a compound of formula (K) with a reductant under conditions sufficient to form a compound of formula (J);

wherein R^4 is C_{1-3} alkyl and each R^5 is independently an optionally substituted C_{1-6} alkyl or an optionally substituted C_{1-6} aryl.

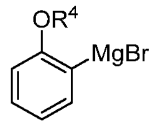
34. A method for preparing compound of formula (J):



(J),

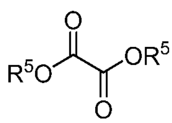
comprising the steps of:

(a) contacting a compound of formula (R):



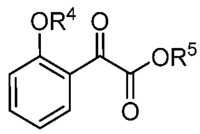
(R)

with a compound of formula (S):



(S)

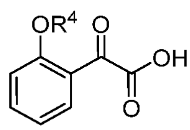
under conditions sufficient to form a compound of formula (P):



(P)

or a solvate or a hydrate thereof,

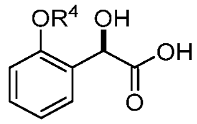
(b) contacting a compound of formula (P), or a solvate or a hydrate thereof, with a base under conditions sufficient to form a compound of formula (O):



(O)

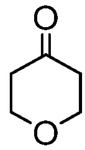
or a salt, a solvate, or a hydrate thereof,

(c) contacting a compound of formula (O), or a salt, a solvate, or a hydrate thereof, with a reductant and a catalyst under conditions sufficient to form a compound of formula (N):



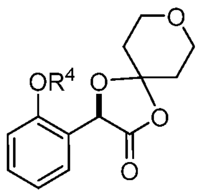
(N);

(d) contacting a compound of formula (N) with a compound of formula (M):



(M)

under conditions sufficient to form a compound of formula (L):

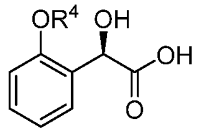


(L);

and (g) contacting a compound of formula (L) with a reductant under conditions sufficient to form a compound of formula (J),

wherein R^4 is C_{1-3} alkyl and each R^5 is independently an optionally substituted C_{1-6} alkyl or an optionally substituted C_{1-6} aryl.

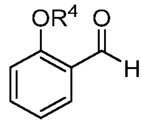
35. A method for preparing a compound of formula (N):



(N),

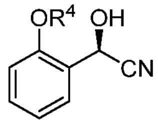
wherein R^4 is C_{1-3} alkyl, comprising:

(a) contacting a compound of formula (U):



(U)

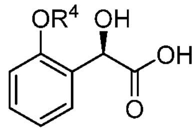
with a hydroxynitrilase and a hydrogen cyanide source under conditions sufficient to form a compound of formula (T):



(T)

and (b) contacting a compound of formula (T) with a nitrilase under conditions sufficient to form a compound of formula (N).

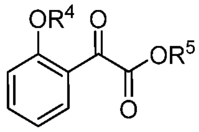
36. A method for preparing a compound of formula (N):



(N),

comprising:

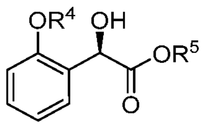
(a) contacting a compound of formula (P):



(P)

or a solvate or a hydrate thereof,

with a ketoreductase under conditions sufficient to form a compound of formula (V):

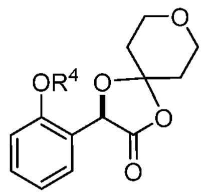


(V)

and (b) contacting a compound of formula (V) with a base under conditions sufficient to form a compound of formula (N),

wherein R^4 is C_{1-3} alkyl and R^5 is an optionally substituted C_{1-6} alkyl or an optionally substituted C_{1-6} aryl.

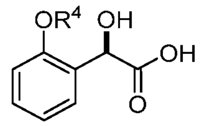
37. A method of preparing a compound of formula (L):



(L),

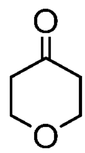
wherein R^4 is C_{1-3} alkyl,

comprising contacting a compound of formula (N):



(N),

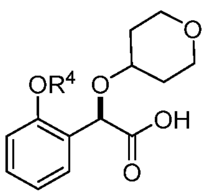
with a compound of formula (M):



(M),

under conditions sufficient to form a compound of formula (L).

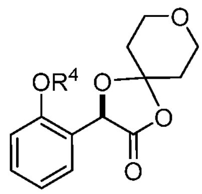
38. A method of preparing a compound of formula (K):



(K),

wherein R^4 is C_{1-3} alkyl,

comprising contacting a compound of formula (L):



(L),

with a reductant under conditions sufficient to form a compound of formula (K).

39. The method of any one of claims 33-38, wherein R^4 is methyl.

40. The method of any one of claims 33-34 and 36, wherein R^5 is ethyl.